

MAT 320 HW09

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Problem 1. Show that $f: (1, +\infty) \rightarrow \mathbb{R}$ defined by $f(x) = \log x$ is uniformly continuous.

Solution. We first prove the following inequality:

Claim 1.1 — For all $x \geq y > 1$,

$$\frac{e^y}{y} \leq \frac{e^x}{x}.$$

Proof. Since $\frac{1}{y} - \frac{1}{x} = \frac{x-y}{xy} \leq x - y$,

$$\begin{aligned} \frac{1}{y} - \frac{1}{x} &\leq x - y \leq (e - 1)(x - y) \\ &= \sum_{k=1}^{\infty} \frac{x - y}{k!} = \sum_{k=2}^{\infty} \frac{(x - y)k}{k!} \\ &\leq \sum_{k=2}^{\infty} \frac{(x - y)(x^{k-2} + x^{k-3}y + \dots + xy^{k-3} + y^{k-2})}{k!} \\ &= \sum_{k=2}^{\infty} \frac{x^{k-1} - y^{k-1}}{k!} \\ &= \left(\frac{e^x}{x} - 1 - \frac{1}{x} \right) - \left(\frac{e^y}{y} - 1 - \frac{1}{y} \right), \end{aligned}$$

which simplifies directly to the inequality we want. $\square$

Now, $\log$ is increasing (because its inverse $\exp$ is also increasing), so we can take logs to get

$$y - \log y \leq x - \log x \iff \log x - \log y \leq x - y.$$

This implies uniform continuity, since $x - y < \varepsilon$ implies $\log x - \log y < \varepsilon$ for all $\varepsilon > 0$. $\square$

Problem 2. Let (X, d) be a metric space and E be a subset of X . Define

$$f_E(x) = \inf_{y \in E} d(x, y).$$

Show that

- (a) f_E is uniformly continuous.
- (b) $f_E(x) = 0$ if and only if $x \in \overline{E}$.
- (c) if A and B are disjoint closed sets then

$$f(x) = \frac{f_A(x)}{f_A(x) + f_B(x)}$$

is a continuous function such that $f = 0$ precisely on the set A and $f = 1$ precisely on the set B .

Solution. Let a, b be in X and let $\varepsilon > 0$. Suppose $d(a, b) < \varepsilon$.

- If one of a and b is in E (say WLOG b), then $f_E(b) = 0$ because it is nonnegative and bounded above by $d(b, b) = 0$, so

$$|f_E(a) - f_E(b)| = f_E(a) = \inf_{y \in E} d(a, y) \leq d(a, b) < \varepsilon,$$

where the first inequality comes from $b \in E$.

- If neither a nor b is in E , then let $\delta > 0$. Then, we can pick x such that

$$d(b, x) \leq f_E(b) + \delta.$$

This lets us bound

$$\begin{aligned} f_E(a) - f_E(b) &= (f_E(a) + \delta) - (f_E(b) + \delta) \\ &\leq d(a, x) - d(b, x) + \delta \\ &\leq d(a, b) + \delta. \end{aligned}$$

This same bound holds for $f_E(b) - f_E(a)$ by symmetry, so, taking δ arbitrarily small, we can conclude that

$$|f_E(a) - f_E(b)| \leq d(a, b) < \varepsilon.$$

Thus, for any a and b ,

$$d(a, b) < \varepsilon \implies |f_E(a) - f_E(b)| < \varepsilon,$$

so f_E is uniformly continuous, proving part (a).

For part (b), suppose first that x is a limit point of E (so it is in $\overline{E}$). Then, for each $\varepsilon > 0$, there is a $y \in E$ such that $d(x, y) < \varepsilon$; by definition of $\inf$, this means that

$$f_E(x) = \inf_{y \in E} d(x, y) = 0.$$

Conversely, if x isn't a limit point of E , then some open ball $B_X(x, \varepsilon)$ contains no points of E ; that is, every point y of E satisfies $d(x, y) \geq \varepsilon$, hence

$$f_E(x) = \inf_{y \in E} d(x, y) \geq \varepsilon > 0.$$

For part (c), we note that A and B being closed means $A = \overline{A}$ and $B = \overline{B}$. Since the sets are disjoint, there is no point at which $f_A(x) = f_B(x) = 0$, ensuring that f is well-defined. f must then be continuous because sums and quotients of continuous functions are continuous as long as the divisor of the quotient is nonzero. Moreover, since f_A and f_B are always nonnegative,

$$0 \leq \frac{f_A(x)}{f_A(x) + f_B(x)} \leq 1.$$

The upper bound is achieved if and only if $f_B(x) = 0$, i.e. when $x \in B$, and the lower bound is achieved if and only if $f_A(x) = 0$, i.e. when $x \in A$. $\square$

Problem 3. Let (X, d) be a metric space and $D \subseteq X$ a dense subset. Show that if $f: D \rightarrow \mathbb{R}$ is uniformly continuous then there exists a unique continuous function $g: X \rightarrow \mathbb{R}$ such that $g(x) = f(x)$ for all $x \in D$.

Solution. First, we show that such a function g exists.

Claim 3.1 — Let (x_n) be a sequence of points in D that converges in X . Then, the sequence $(f(x_n))$ converges in $\mathbb{R}$.

Proof. Note that (x_n) being a convergent sequence means that it is Cauchy, as shown in Problem 6. By uniform continuity, we know that, for each $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$d(x_m, x_n) < \delta \implies |f(x_m) - f(x_n)| < \varepsilon.$$

Since (x_n) is Cauchy, there exists N such that $m, n > N$ implies that $d(x_m, x_n) < \delta$, so

$$m, n > N \implies d(x_m, x_n) < \delta \implies |f(x_m) - f(x_n)| < \varepsilon,$$

which tells us that $(f(x_n))$ is a Cauchy sequence, so it converges because $\mathbb{R}$ is complete. $\square$

Claim 3.2 — Let (Y, d_Y) be a metric space. If (a_n) is a sequence of points in Y with limit ℓ and (b_n) is a sequence of points in Y such that $d_Y(a_n, b_n) \rightarrow 0$, then

$$\lim_{n \rightarrow \infty} b_n = \ell.$$

Proof. For each $\varepsilon > 0$, there exists N such that

$$n > N \implies d_Y(\ell, a_n) < \frac{\varepsilon}{2}, \quad d_Y(a_n, b_n) < \frac{\varepsilon}{2},$$

so, for $n > N$,

$$d_Y(\ell, b_n) \leq d_Y(\ell, a_n) + d_Y(a_n, b_n) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which is what it means for b_n to converge to ℓ . $\square$

Claim 3.3 — Let (x_n) and (y_n) be two sequences of points in D that converge to the same point $\ell \in X$. Then,

$$\lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} f(y_n).$$

Proof. Let $\varepsilon > 0$. We know that there exists $\delta > 0$ such that

$$d(x_n, y_n) < \delta \implies |f(x_n) - f(y_n)| < \varepsilon$$

because f is uniformly continuous. Moreover, since $x_n \rightarrow \ell$ and $y_n \rightarrow \ell$,

$$d(x_n, \ell) < \frac{\delta}{2}, \quad d(y_n, \ell) < \frac{\delta}{2}$$

are true eventually, say when $n > N$, so

$$d(x_n, y_n) < \delta$$

is true for all $n > N$. That is,

$$n > N \implies d(x_n, y_n) < \delta \implies |f(x_n) - f(y_n)| < \varepsilon.$$

Thus,

$$\lim_{n \rightarrow \infty} f(x_n) - f(y_n) = 0 \implies \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} f(y_n)$$

with the implication coming from Claim 3.2 (the limits both exist by Claim 3.1). $\square$

Therefore, for any point $x \in X$, we can define

$$g(x) := \lim_{n \rightarrow \infty} f(x_n)$$

where (x_n) is any sequence of points in D that converges to x . Such a sequence (x_n) exists because we can take x_n to be any point in $D \cap B(x, \frac{1}{n})$, which is nonempty because D is dense in X . Also, $g(x)$ well-defined because Claim 3.1 says that the limit exists and Claim 3.3 says that the value of the limit is only dependent on x .

We now have to show that $g(x) = f(x)$ for all $x \in D$ and g is continuous. For the first part, we just observe that if $x \in D$ then we can take $x_n = x$ for all n so that

$$g(x) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} f(x) = f(x).$$

To show that g is continuous at any $c \in X$, let (x_n) be a sequence of points in X converging to c . By density we can pick a sequence (y_n) of points in D such that $d(x_n, y_n) < \frac{1}{n}$. Since

$$d(y_n, c) \leq d(y_n, x_n) + d(x_n, c)$$

and the right-hand side has limit zero, we can say that $d(y_n, c) \rightarrow 0$, implying $y_n \rightarrow c$. Letting $\varepsilon > 0$, uniform continuity says that there exists a δ such that

$$d(x_n, y_n) < \delta \implies |f(x_n) - f(y_n)| < \varepsilon,$$

so the bound on $|f(x_n) - f(y_n)|$ holds as long as $n > \frac{1}{\delta}$. That is, for any $\varepsilon > 0$, we have $|f(x_n) - f(y_n)| < \varepsilon$ eventually, which lets us say that

$$\lim_{n \rightarrow \infty} f(x_n) - f(y_n) = 0 \implies \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} f(y_n) = g(c),$$

again using Claim 3.2 to deduce that the limits are equal.

Now, we show that g is unique. Indeed, suppose that $h: X \rightarrow \mathbb{R}$ is another continuous function such that $h(x) = f(x)$ for all $x \in D$. Then, $g - h$ is a continuous function that is zero on all of D . Now, for any $x \in X$, let $x_n \in D$ be a point such that $d(x, x_n) < \frac{1}{n}$; these x_n exist because D is dense in X . By continuity,

$$g(x) - h(x) = \lim_{n \rightarrow \infty} g(x_n) - h(x_n) = \lim_{n \rightarrow \infty} 0 = 0,$$

meaning $g(x) = h(x)$ for all x , i.e. g and h are the same function. $\square$

Problem 4. Let (X, d) be a metric space. Show that if each closed and bounded subset of X is compact, then (X, d) is complete.

Solution. Let (x_n) be an arbitrary Cauchy sequence in X . We showed in class that $S = \{x_n\}$ is bounded, so the closure of this set is bounded:

Claim 4.1 — For a set C with finite diameter, $\text{diam } C = \text{diam } \overline{C}$.

Proof. Since $C \subseteq \overline{C}$,

$$\text{diam } C \leq \text{diam } \overline{C}.$$

To show the opposite inequality, we prove equivalently that

$$\text{diam } \overline{C} \leq \text{diam } C + \varepsilon$$

for every $\varepsilon > 0$. Indeed, let $x, y \in \overline{C}$. Then, x and y are limit points of C , so we can find $x', y' \in C$ such that $d(x, x')$ and $d(y, y')$ are both less than $\frac{\varepsilon}{2}$. The triangle inequality then says that

$$d(x, y) \leq d(x, x') + d(x', y') + d(y', y) < \frac{\varepsilon}{2} + \text{diam } C + \frac{\varepsilon}{2} = \text{diam } C + \varepsilon.$$

Thus $\text{diam } C + \varepsilon$ is an upper bound for $d(x, y)$, meaning it is at least as large as $\text{diam } C$, as desired. $\square$

This means that $\overline{S}$ is compact because it is a closed and bounded subset of X . Now, $\overline{S}$ is sequentially compact, so (x_n) has a convergent subsequence (x_{n_k}) with limit ℓ . I claim that (x_n) converges to ℓ as well. Let $\varepsilon > 0$ and pick N so that

$$\begin{aligned} n_k > N &\implies |\ell - x_{n_k}| < \frac{\varepsilon}{2} \\ m, n > N &\implies |x_m - x_n| < \frac{\varepsilon}{2}, \end{aligned}$$

where the latter inequality comes from (x_n) being Cauchy. Then, if n_k and n are both bigger than N ,

$$|\ell - x_n| \leq |\ell - x_{n_k}| + |x_{n_k} - x_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

proving that $x_n \rightarrow \ell$, i.e. every Cauchy sequence in X converges. $\square$

Problem 5. Let (X, d) be a metric space. Show that

- (a) if $E \subseteq X$ is complete in the induced metric then E is a closed subset of X .
- (b) if X is complete, then a subset $E \subseteq X$ is closed if and only if it is complete in the induced metric.

Solution. For (a), let ℓ be a limit point of E . For every positive integer n , there exists an x_n such that $d(\ell, x_n) < \frac{1}{n}$. Then, the sequence (x_n) converges to ℓ by construction, so, by Problem 6, it is Cauchy. By completeness, (x_n) must therefore converge to a point in E , so ℓ , the limit of x_n , is in E . Thus E is closed because it contains all of its limit points.

For (b), we already showed that complete implies closed. Conversely, if E is closed, consider a Cauchy sequence (x_n) of points in E . Since this sequence is also Cauchy in X , it has a limit ℓ . However, ℓ is then a limit point of E , so E being closed means that $\ell \in E$. Hence, every Cauchy sequence of points in E has a limit in E , i.e. E is complete. $\square$

Problem 6. Show that if $x_n \rightarrow \ell$ in a metric space (X, d) , then (x_n) is a Cauchy sequence.

Solution. Let $\varepsilon > 0$. Convergence to ℓ means that there exists an N such that

$$n > N \implies |x_n - \ell| < \frac{\varepsilon}{2}.$$

Then,

$$m, n > N \implies |x_m - x_n| \leq |x_m - \ell| + |\ell - x_n| < \varepsilon,$$

so the sequence is Cauchy. $\square$

Problem 7. Let (X, d_X) be a metric space and S a set. We say that a function $f: S \rightarrow X$ is bounded if its image $f(S) \subseteq X$ has finite diameter. Let

$$\mathcal{B}(S, X) = \{f: S \rightarrow X : f \text{ is bounded}\}.$$

Show that

(a) if f and g are bounded then

$$\sup_{s \in S} d_X(f(s), g(s)) < +\infty.$$

(b) the function

$$d(f, g) = \sup_{s \in S} d_X(f(s), g(s))$$

is a metric on $\mathcal{B}(S, X)$.

(c) if X is complete, then $(\mathcal{B}(S, X), d)$ is a complete metric space.

Solution. (a) Fix $a \in S$. Then, for any s ,

$$\begin{aligned} d_X(f(s), g(s)) &\leq d_X(f(s), f(a)) + d_X(f(a), g(a)) + d_X(g(a), g(s)) \\ &\leq d_X(f(a), g(a)) + \text{diam}(f(S)) + \text{diam}(g(S)), \end{aligned}$$

so $d_X(f(s), g(s))$ is bounded above by a constant, hence its supremum over all s is real.

(b) Moreover, d is a metric. First, it is clearly nonnegative because it is the supremum of a collection of nonnegative values. Second, it is zero iff f and g are the same function because

$$\sup_{s \in S} d_X(f(s), g(s)) = 0 \implies d_X(f(s), g(s)) \leq 0 \text{ for all } s \in S$$

which implies $d_X(f(s), g(s))$ is identically zero and so $f(s) = g(s)$ for all $s \in S$, and

$$d(f, f) = \sup_{s \in S} d_X(f(s), f(s)) = \sup_{s \in S} 0 = 0.$$

It is also trivial to see that $d(f, g) = d(g, f)$. Finally, the triangle inequality holds because

$$\begin{aligned} d(f, h) + d(g, h) &= \sup_{s \in S} d_X(f(s), h(s)) + \sup_{s \in S} d_X(g(s), h(s)) \\ &= \sup_{s \in S} [d_X(f(s), h(s)) + d_X(g(s), h(s))] \\ &\geq \sup_{s \in S} d_X(f(s), g(s)) = d(f, g). \end{aligned}$$

(c) Lastly, suppose X is complete. Then, let (f_n) be a Cauchy sequence in $\mathcal{B}(S, X)$. We see that, for each $s \in S$, the sequence $(f_n(s))$ is Cauchy in X because, for $\varepsilon > 0$, if N is chosen such that $m, n > N$ implies $d(f_m, f_n) < \varepsilon$, then

$$d_X(f_m(s), f_n(s)) \leq d(f_m, f_n) < \varepsilon.$$

Thus $(f_n(s))$ converges in X because X is complete, so (f_n) has a pointwise limit f .

I claim that f is also the limit of (f_n) with respect to d . Let $\varepsilon > 0$ and take N such that $d(f_m, f_n) < \varepsilon/2$ whenever m and n exceed N . Then, for any $s \in S$, $m, n > N$ implies

$$\begin{aligned} d_X(f_n(s), f(s)) &\leq d_X(f_n(s), f_m(s)) + d_X(f_m(s), f(s)) \\ &\leq d(f_n, f_m) + d_X(f_m(s), f(s)) \\ &< \varepsilon/2 + d_X(f_m(s), f(s)). \end{aligned}$$

Taking $m \rightarrow \infty$, this means that

$$d_X(f_n(s), f(s)) \leq \varepsilon/2 < \varepsilon$$

for all $s \in S$, so for any $\varepsilon > 0$ there is an N such that $n > N$ implies that $d(f_n, f) < \varepsilon$, which is what it means for f_n to converge to f with respect to the metric d . $\square$

Problem 8. Let (K, d_K) and (X, d_X) be two metric spaces and $C^0(K, X)$ be the space of all continuous functions from K to X . Show that if K is compact and X is complete then $C^0(K, X)$ is a complete metric space with metric

$$d(f, g) = \max_{x \in K} d_X(f(x), g(x)).$$

Solution. First, we show that d is well-defined; that is,

$$\max_{x \in K} d_X(f(x), g(x))$$

actually exists for each f and g .

Claim 8.1 — For any continuous functions $f, g: K \rightarrow X$, the function $\alpha: K \rightarrow \mathbb{R}$ defined by

$$\alpha(x) = d_X(f(x), g(x))$$

is continuous.

Proof. Let c be an element of X and let $\varepsilon > 0$. Then,

$$\begin{aligned} \alpha(c) &\leq d_X(f(c), f(x)) + \alpha(x) + d_X(g(x), g(c)), \\ \alpha(x) &\leq d_X(f(c), f(x)) + \alpha(c) + d_X(g(x), g(c)). \end{aligned}$$

This means that we can bound

$$|\alpha(x) - \alpha(c)| \leq d_X(f(c), f(x)) + d_X(g(x), g(c)).$$

By continuity of f and g , we can pick $\delta > 0$ so that

$$d_K(x, c) < \delta \implies d_X(f(x), f(c)) < \varepsilon/3, \quad d_X(g(x), g(c)) < \varepsilon/3.$$

This means that $d_K(x, c) < \delta$ implies

$$|\alpha(x) - \alpha(c)| \leq \frac{2\varepsilon}{3} < \varepsilon,$$

which is what it means for α to be continuous. $\square$

Now the Weierstrass theorem implies that $\alpha(K) \subseteq \mathbb{R}$ is an interval, so it has a maximum. To show that d is a metric, we observe the following facts:

- $d(f, g)$ is nonnegative because it is the maximum of a collection of nonnegative values.
- $d(f, g) = 0$ if $f = g$ because

$$d(f, f) = \max_{x \in K} d_X(f(x), g(x)) = \max_{x \in K} 0 = 0,$$

and conversely, if $d(f, g) = 0$, then

$$0 \leq d_X(f(x), g(x)) \leq 0$$

for all $x \in K$, which forces $d_X(f(x), g(x)) = 0 \iff f(x) = g(x)$ on all of K .

- $d(f, g) = d(g, f)$ is trivial, as $\max(a, b) = \max(b, a)$.
- The triangle inequality holds because

$$\begin{aligned} d(f, h) + d(g, h) &= \max_{x \in K} d_X(f(x), h(x)) + \max_{x \in K} d_X(g(x), h(x)) \\ &= \max_{x \in K} [d_X(f(x), h(x)) + d_X(g(x), h(x))] \\ &\geq \max_{x \in K} d_X(f(x), g(x)) = d(f, g). \end{aligned}$$

It remains to prove that $(C^0(K, X), d)$ is complete, which can be done in the same way as 7(c). Let (f_n) be a Cauchy sequence in $C^0(K, X)$. We see that, for each $x \in K$, the sequence $(f_n(x))$ is Cauchy in X because, for $\varepsilon > 0$, if N is chosen such that $m, n > N$ implies $d(f_m, f_n) < \varepsilon$, then

$$d_X(f_m(x), f_n(x)) \leq d(f_m, f_n) < \varepsilon.$$

Thus $(f_n(x))$ converges in X because X is complete, so (f_n) has a pointwise limit f .

I claim that f is also the limit of (f_n) with respect to d . Let $\varepsilon > 0$ and take N such that $d(f_m, f_n) < \varepsilon/2$ whenever m and n exceed N . Then, for any $x \in K$, $m, n > N$ implies

$$\begin{aligned} d_X(f_n(x), f(x)) &\leq d_X(f_n(x), f_m(x)) + d_X(f_m(x), f(x)) \\ &\leq d(f_n, f_m) + d_X(f_m(x), f(x)) \\ &< \varepsilon/2 + d_X(f_m(x), f(x)). \end{aligned}$$

Taking $m \rightarrow \infty$, this means that

$$d_X(f_n(x), f(x)) \leq \varepsilon/2 < \varepsilon$$

for all $x \in K$, so for any $\varepsilon > 0$ there is an N such that $n > N$ implies that $d(f_n, f) < \varepsilon$, which is what it means for f_n to converge to f with respect to the metric d . $\square$

Problem 9. Prove that ℓ^1 is a complete metric space.

Solution. Let $(\mathbf{x}_n)$ be a Cauchy sequence in ℓ^1 . Then, for each $\varepsilon > 0$, there exists N such that $m, n > N$ implies

$$\sum_{k=1}^{\infty} |x_{m,k} - x_{n,k}| < \varepsilon.$$

Since $|x_{m,k} - x_{n,k}| < \|\mathbf{x}_m - \mathbf{x}_n\|$ (where the norm is $\|\mathbf{x}\| = d(\mathbf{x}, \mathbf{0})$), this means the sequence $(x_{n,k})_{n \in \mathbb{N}}$ of the k th coordinates of each $\mathbf{x}_n$ is Cauchy, so, since $\mathbb{R}$ is complete, $(\mathbf{x}_n)$ converges coordinatewise (i.e. $(x_{n,k})_n$ converges for every k). Let ℓ_k be the limit of $x_{n,k}$ as $n \rightarrow \infty$.

Claim 9.1 — The vector

$$\mathbf{x} = (\ell_1, \ell_2, \ell_3, \dots)$$

is in ℓ^1 .

Proof. We wish to show that

$$\sum_{i=1}^{\infty} |\ell_i|$$

converges. It suffices to show that the partial sums are bounded because the numbers being summed are all nonnegative.

For each $i \leq N$, there exists an M_i such that

$$n > M_i \implies |\ell_i - x_{n,i}| < 2^{-i}.$$

Fix $n > \max\{M_1, \dots, M_N\}$. The N th partial sum can be bounded above as follows:

$$\begin{aligned} \sum_{i=1}^N |\ell_i| &\leq \sum_{i=1}^N |\ell_i - x_{n,i}| + \sum_{i=1}^N |x_{n,i}| \\ &< \sum_{i=1}^N 2^{-i} + \|\mathbf{x}_n\| \\ &< 1 + \|\mathbf{x}_n\|. \end{aligned}$$

Additionally, since Cauchy sequences are bounded, we can bound $\|\mathbf{x}_n\|$ above by a constant C for all n , as, for an arbitrary fixed r , we can furnish a constant upper bound in the form

$$\|\mathbf{x}_n\| \leq \|\mathbf{x}_n - \mathbf{x}_r\| + \|\mathbf{x}_r\| \leq \text{diam}\{\mathbf{x}_n\} + \|\mathbf{x}_r\| =: C.$$

Therefore we have bounded the partial sums of the $|\ell_i|$ by the constant $1 + C$, so the infinite series converges. $\square$

Now, we just need to prove that $\mathbf{x}_n \rightarrow \ell$ in the metric of ℓ^1 . This is done with the same idea of bounding partial sums. Let $\varepsilon > 0$ and $N \in \mathbb{N}$. For each $i \in \{1, 2, \dots, N\}$, we can find thresholds M_i for which

$$m > M_i \implies |\ell - x_{m,i}| < \varepsilon \cdot 2^{-i-1}.$$

This means that, if $m > \max\{M_1, \dots, M_N\}$, then

$$\sum_{i=1}^N |\ell_i - x_{m,i}| < \frac{\varepsilon}{2}.$$

We also know that $(\mathbf{x}_n)$ being Cauchy means that there is an M such that

$$m, n > M \implies \|\mathbf{x}_m - \mathbf{x}_n\| < \frac{\varepsilon}{2}.$$

Putting this together, we can fix m to be larger than M and all of the M_i and use the triangle inequality to say that $n > M$ implies

$$\begin{aligned} \sum_{i=1}^N |\ell_i - x_{n,i}| &\leq \sum_{i=1}^N |\ell_i - x_{m,i}| + \sum_{i=1}^N |x_{m,i} - x_{n,i}| \\ &\leq \sum_{i=1}^N \varepsilon \cdot 2^{-i-1} + \sum_{i=1}^{\infty} |x_{m,i} - x_{n,i}| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

In other words, as long as $n > M$ (noting that M only depends on ε), we can bound all the partial sums of N below ε , so $n > M$ implies that

$$\|\mathbf{x} - \mathbf{x}_n\| < \varepsilon,$$

which is what it means for $\mathbf{x}_n$ to converge to $\mathbf{x}$ in the metric of ℓ^1 , so we are done. $\square$

Problem 10. Prove that $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly on a set $E \subseteq \mathbb{R}$ then $f_n + g_n \rightarrow f + g$ uniformly on E . Show an example where $f_n \rightarrow f$ and $g_n \rightarrow g$ on some set $E \subseteq \mathbb{R}$ but it is not true that $f_n \cdot g_n \rightarrow f \cdot g$ on $E \subseteq \mathbb{R}$.

Solution. If $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly on $E \subseteq \mathbb{R}$, then we can say that for each $\varepsilon > 0$, there exists an N such that $n > N$ implies

$$|f(x) - f_n(x)| < \frac{\varepsilon}{2}, \quad |g(x) - g_n(x)| < \frac{\varepsilon}{2}$$

for all $x \in E$. For such n ,

$$|f(x) + g(x) - f_n(x) - g_n(x)| \leq |f(x) - f_n(x)| + |g(x) - g_n(x)| < \varepsilon.$$

Again, noting that this holds for all x , we can conclude that $f_n + g_n \rightarrow f + g$ uniformly.

For products: let $E = \mathbb{R}$ and $f_n(x) = g_n(x) = x + \frac{1}{n}$ for all x . Then, $f(x) = x$ and $g(x) = x$ are the uniform limits of f_n and g_n respectively because

$$|f(x) - f_n(x)| = |g(x) - g_n(x)| = \frac{1}{n} < \varepsilon$$

whenever $n > \frac{1}{\varepsilon}$, and the threshold for n is independent of x . On the other hand,

$$f_n(x) \cdot g_n(x) = \left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2}, \quad f(x)g(x) = x^2.$$

In this case, for a fixed n ,

$$|f(x)g(x) - f_n(x)g_n(x)| = \left| \frac{2x}{n} + \frac{1}{n^2} \right|$$

can be arbitrarily large, so any constant upper bound ε cannot hold for all x ; therefore, $f_n g_n$ does not converge to $f g$ uniformly. $\square$

Problem 11. Suppose that $f_n \rightarrow f$ uniformly on an interval (a, b) . Prove that if the f_n are uniformly continuous functions then so is their uniform limit f .

Solution. Let the codomain of the f_n be the metric space (X, d) and let $\varepsilon > 0$. Note that, for any $x, y \in (a, b)$,

$$d(f(x), f(y)) \leq d(f(x), f_n(x)) + d(f_n(x), f_n(y)) + d(f_n(y), f(y)).$$

Since $f_n \rightarrow f$ uniformly, there exists an N such that $n > N$ implies

$$d(f(z), f_n(z)) < \frac{\varepsilon}{3}$$

for all $z \in X$, so fix n to be an integer greater than N . Now, f_n is uniformly continuous, there exists a $\delta > 0$ such that

$$|x - y| < \delta \implies d(f_n(x), f_n(y)) < \frac{\varepsilon}{3}.$$

Thus, for each $\varepsilon > 0$, there exists a $\delta > 0$ for which

$$|x - y| < \delta \implies d(f(x), f(y)) < \varepsilon,$$

which is what it means for f to be uniformly continuous. □